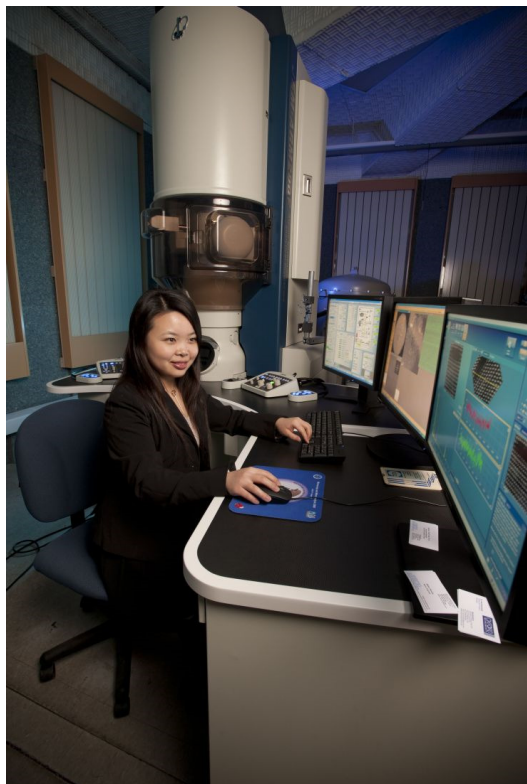


JEOL JEM-ARM200CF

Aberration Corrected Cold Field Emission Scanning Transmission Electron Microscope

Location: Science and Engineering South, 104A



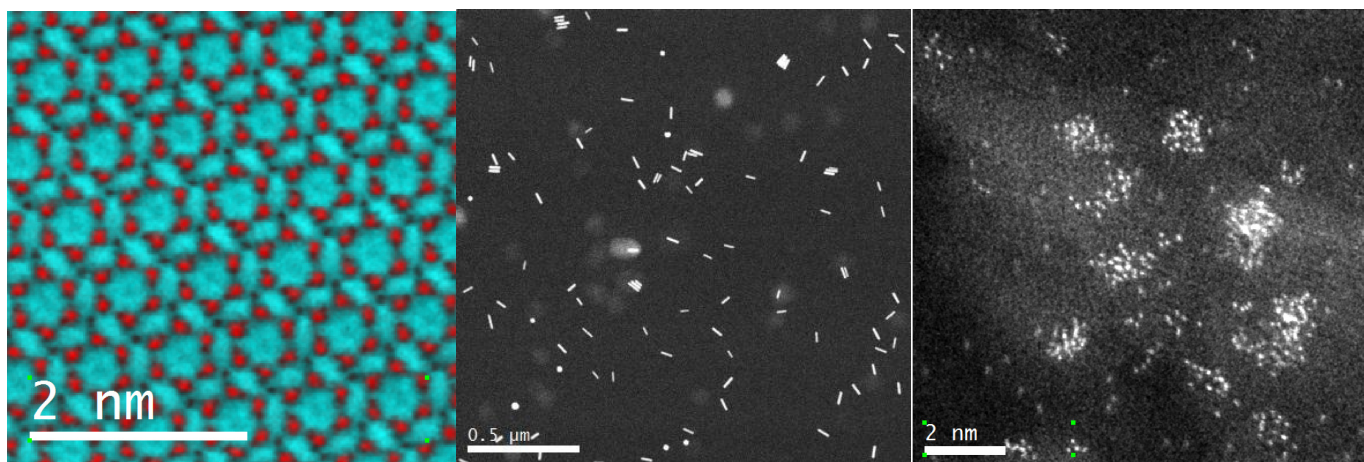
Description

The JEM-ARM200CF is a probe aberration corrected 200kV STEM/TEM with a cold field emission source with 0.35eV energy resolution. For HAADF imaging at 200kV this instrument has a resolution of less than 0.08nm. It was installed in 2011

There are HAADF, LAADF, BF/ABF and a Gatan ADF/BF STEM detector. Up to four STEM images can be recorded simultaneously. For TEM Imaging the microscope has two CCD cameras - an upper 4MP camera and a lower 11MP camera. For microanalysis the microscope is equipped with a Gatan Quatum GIF and an Oxford X-max 100TLE windowless SDD X-ray detector both of which are capable of atomic resolution mapping.

In addition to the standard holders, Double Tilt Heating (1000°C), Double Tilt Cooling (-170°C), STM-TEM, Double Tilt Electrothermal (1200°C), Liquid, Vacuum Transfer and Single Tilt Tomography (+/- 70 deg) holders are available for this microscope.

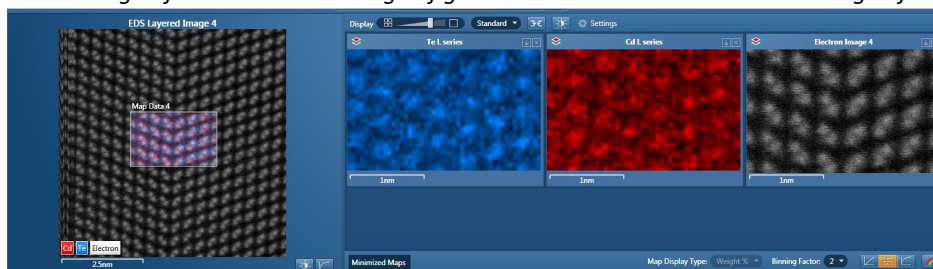
Reductions in quoted unassisted rates are offered to groups that use more than 60 hours of aberration corrected microscope time in a calendar month.



Overlay HAADF and ABF image of Si₃N₄

HAADF image of gold nanorods in water

HAADF image of Pt atoms on Alumina



Atomic resolution XEDS maps of Te (blue) and Cd (red) across a twin boundary

JEOL JEM-ARM200CF

Technical Specifications

- STEM resolution (HAADF): 0.078nm @ 200kV; 0.136nm @ 80kV
- STEM Resolution (BF) 0.136nm @ 200kV
- TEM Resolution: 0.10nm lattice, 0.19nm point-to-point @ 200kV.
- Accelerating Voltage: 200, 120, 80kV.
- Objective lens (TEM): focal length 1.9mm, Cs 0.5mm, Cc 1.1mm, minimum focus step 0.25nm.
- Objective Lens (STEM) Cs -0.1mm to 0.6mm, Cc 1.4mm
- Magnification Range (TEM): 2,000x - 2,000,000x.
- Magnification Range (STEM) 20,000x - 150,000,000 x
- Camera length range: 80 - 2,000mm.
- Specimen Tilt range up to +/-25 degrees (X & Y) depending on Specimen Stage.
- Specimen movement: 2mm (X,Y), 0.1mm (Z)
- Energy Resolution (Quantum): 0.35eV